

Oxygen-Vacancy-Enhanced Peroxidase-like Activity of Reduced Co_3O_4 Nanocomposites for the Colorimetric Detection of H_2O_2 and Glucose

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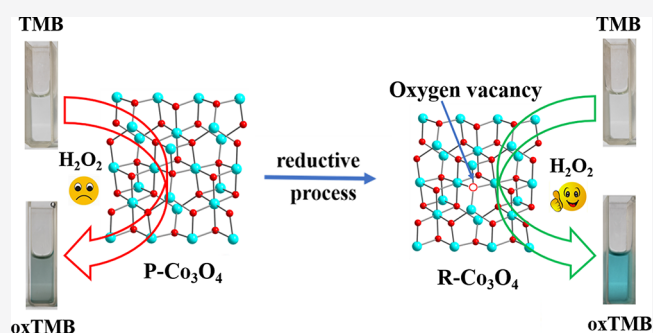
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ABSTRACT: Colorimetric assays have drawn increasing research interest with respect to the quantitative detection of hydrogen peroxide (H_2O_2) based on artificial enzymes because of their advantages with respect to natural enzymes, including design flexibility, low cost, and high stability. Regardless, the majority of the artificial enzymes exhibit low affinity to H_2O_2 with large Michaelis–Menten constants (K_m). This indicates that the catalytic oxidation of 3,3',5,5'-tetramethylbenzidine (TMB) to blue-colored oxTMB requires a high H_2O_2 concentration, hindering the sensitivity of the colorimetric assay. To address this problem, novel reduced Co_3O_4 nanoparticles ($\text{R-Co}_3\text{O}_4$) have been synthesized in this study via a step-by-step procedure using ZIF-67 as the precursor. $\text{R-Co}_3\text{O}_4$ exhibits a considerably enhanced peroxidase-like activity when compared with that exhibited by pristine Co_3O_4 ($\text{P-Co}_3\text{O}_4$). The catalytic process in the case of $\text{R-Co}_3\text{O}_4$ occurs in accordance with the typical Michaelis–Menten equation, and the affinity of $\text{R-Co}_3\text{O}_4$ to H_2O_2 is apparently higher than that of $\text{P-Co}_3\text{O}_4$. Furthermore, the density functional theory calculations revealed that the introduction of oxygen vacancies to $\text{R-Co}_3\text{O}_4$ enhances its H_2O_2 adsorption ability and facilitates the decomposition of H_2O_2 to produce $\cdot\text{OH}$ radicals, resulting in improved peroxidase-like activity. A simple and convenient colorimetric assay has been established based on the excellent peroxidase-like activity of $\text{R-Co}_3\text{O}_4$ for detecting H_2O_2 in concentrations of 1–30 μM with a detection limit of 4.3×10^{-7} mol/L ($S/N = 3$). Furthermore, the $\text{R-Co}_3\text{O}_4$ -based colorimetric method was successfully applied to glucose detection in human serum samples, demonstrating its potential for application in complex biological systems.



1. INTRODUCTION

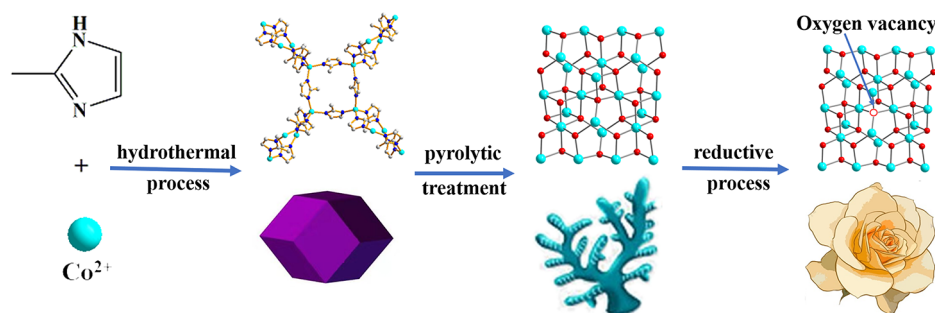
As is well-known, hydrogen peroxide (H_2O_2) is associated with cell damage and several conditions, including autoimmune diseases, Alzheimer's disease, and cancer.^{1–5} Thus, the accurate determination of H_2O_2 in complex biological systems is indispensable to conduct clinical research and diagnosis. Several analytical techniques, including electrochemistry, liquid chromatography, chemiluminescence, fluorescence, and colorimetric assays, are currently being used for the quantitative detection of H_2O_2 .^{6–16} Colorimetric assays are gaining increasing research interest because of its ability to quickly provide easily readable results by visual determination via the naked eye or convenient portable instruments.^{17–19}

Natural enzymes can be used to prepare diagnostic kits for the colorimetric detection of H_2O_2 . However, their low stabilities and high cost severely hinder their large-scale application. Artificial enzymes, especially transition metal oxide nanocomposites, are ideal alternatives for natural enzymes owing to their design flexibility, low cost, earth abundance, and high stability under harsh conditions.^{20–22}

Regardless, the majority of the artificial enzymes show a low affinity to H_2O_2 with large Michaelis–Menten constants (K_m). This indicates that the catalytic oxidation of TMB to blue-colored oxTMB using an enzyme, upon which such assays are typically based, requires high H_2O_2 concentrations, decreasing the sensitivity of the colorimetric assay. Therefore, the development of strategies for preparing artificial enzymes with high H_2O_2 affinities is highly desired.

Very recently, Sun et al. investigated the adsorption behavior of H_2O_2 on perfect and oxygen-defective Fe_2O_3 surfaces through the density functional theory (DFT) calculations.²³ Their results denoted that the oxygen vacancies improved the adsorption of H_2O_2 and promoted the decomposition of H_2O_2 on the catalyst surface. Therefore, introducing oxygen

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Scheme 1. Schematic Illustration of the Preparation Process for R-Co₃O₄

vacancies into the transition-metal-oxide nanocomposites could be an effective methodology to enhance their H₂O₂ affinities, resulting in an enhanced peroxidase-like activity for the colorimetric detection of H₂O₂. However, to the best of our knowledge, the effects of oxygen vacancies on the peroxidase-like activity of artificial enzymes have not been yet been studied.

Metal–organic frameworks (MOFs) have been extensively studied as a new class of crystalline porous inorganic–organic hybrid materials owing to their tunable pore sizes, high surface areas, and easy modification of their constituent metal nodes and ligands.^{24–26} MOFs are considered to be ideal candidates for becoming enzyme mimics. However, the majority of MOFs are unstable when exposed to humid environments, severely restricting their applications in this field. Recently, considerable research effort has been devoted to fabricate nanocomposites containing metal oxides embedded in porous carbon using MOFs as the sacrificial templates.^{27–29} During the pyrolysis process, the organic ligands can be transformed into porous carbon, which not only effectively protects the metal oxides from harsh environments and improves the stability of the nanocomposites but also prevents the self-agglomeration of the metal oxide particles and improves their catalytic properties. Thus, the development of strategies that involve the usage of MOFs as precursors for preparing nanocomposites with highly enzymelike activities is highly promising.

Accordingly, in this study, a novel reduced Co₃O₄ nanocomposite (R-Co₃O₄) has been synthesized through a step-by-step procedure using the MOF ZIF-67 as the precursor (Scheme 1). Compared to pristine Co₃O₄ (P-Co₃O₄), R-Co₃O₄ shows a remarkably enhanced peroxidase-like activity. The catalytic process in case of R-Co₃O₄ proceeds in accordance with the typical Michaelis–Menten equation, and the affinity of R-Co₃O₄ to H₂O₂ is apparently higher than that of P-Co₃O₄. Furthermore, the DFT calculations revealed that the introduction of oxygen vacancies to R-Co₃O₄ notably enhances its H₂O₂ adsorption ability and facilitates the decomposition of H₂O₂ to produce ·OH radicals, resulting in an improved peroxidase-like activity. R-Co₃O₄ was successfully used for the colorimetric detection of H₂O₂ and glucose in human serum samples with high sensitivity and selectivity. Thus, this study provides a promising strategy for designing highly sensitive colorimetric sensors intended for performing biological analysis and clinical diagnosis.

2. MATERIALS AND METHODS

2.1. Reagents and Materials. 3,3',5,5'-Tetramethylbenzidine (TMB) and horseradish peroxidase (HRP, 300 units/mg) were purchased from Yuanye Biotechnology (Shanghai) Co., Ltd. Cobalt nitrate (III) hexahydrate [Co(NO₃)₃·6H₂O], 2-methylimidazole,

glucose, and glucose oxidase (GOx) were purchased from Inncochem (Beijing) Co., Ltd. The serum samples were obtained from the Weifang traditional Chinese hospital. All the reagents used in this study were analytical-grade and were used as received without any further purification.

2.2. Characterization. The sample morphologies were characterized by transmission electron microscopy (TEM, JEOL JEM-2010) and field-emission scanning electron microscopy (FE-SEM, JEOL JSM-6700F). X-ray photoelectron spectroscopy (XPS) was conducted using a scanning X-ray microprobe (Thermo Scientific, ESCALAB 250Xi). The electron spin resonance (ESR) spectra were obtained via JEOL JES-FA200 (JEOL, Tokyo, Japan), and the powder X-ray diffraction (PXRD) patterns were collected using a Bruker AXS D₈ Advance instrument. The UV–vis spectra were obtained using the Persee TU-1901 UV–vis spectrophotometer. The N₂ adsorption–desorption measurements were recorded using the ASAP-2020 surface area analyzer at 77 K. Furthermore, the specific surface areas and pore size distributions of the samples were calculated using the Brunauer–Emmett–Teller (BET) method and Barrett–Joyner–Halenda (BJH) model, respectively.

2.3. Synthesis of ZIF-67. ZIF-67 was synthesized using a simple one-step hydrothermal process in accordance with a previous report.³⁰ Typically, 2-methylimidazole (1642 mg, 20.0 mmol) and Co(NO₃)₂·6H₂O (1164 mg, 4.0 mmol) were separately dissolved in 100 mL of methanol. Both the solutions were mixed and stirred for 24 h. The violet precipitate was subsequently collected by filtration, washed thrice using ethanol, and dried overnight at 80 °C.

2.4. Synthesis of P-Co₃O₄ and R-Co₃O₄. The as-prepared ZIF-67 was placed in a porcelain boat and heated to 500 °C at a heating rate of 1 °C min^{−1} for 1 h under an O₂ atmosphere. P-Co₃O₄ was obtained as a black power. Finally, R-Co₃O₄ was obtained by soaking the P-Co₃O₄ powder in 100 mL of 1 M NaBH₄ for 1 h.

2.5. Kinetic Analysis. The kinetic parameters with respect to P-Co₃O₄ and R-Co₃O₄ were explored by varying the substrate concentration from 0.5 to 1.0 mM in 2.0 mL of NaOAc buffer solution (pH 5.0) at 25 °C. The Michaelis–Menten constant (*K_m*) was calculated from the Lineweaver–Burk plots of the double reciprocal of the Michaelis–Menten equation, i.e., 1/*ν* = *K_m*/*V_{max}* (1/[S] + 1/*K_m*), where *ν* denotes the initial velocity, *V_{max}* denotes the maximal reaction velocity, *K_m* denotes the Michaelis–Menten constant, and [S] denotes substrate concentration.

2.6. Detection of H₂O₂ and Glucose Based on the R-Co₃O₄ Nanocomposite. H₂O₂ detection was performed as follows. First, 200 μL of R-Co₃O₄ (0.4 mg/mL), 200 μL of TMB (1 mM), and 200 μL of H₂O₂ (different concentrations) were added to 1400 μL of the NaOAc buffer (0.036 M, pH 4). The resultant solution was used for conducting the absorption spectroscopy measurements after incubation at room temperature for 15 min.

Glucose detection was performed as follows. First, 1000 μL of glucose oxidase (GOx) (0.5 mg/mL) and 1000 μL of the glucose solution (various concentrations) prepared in the NaOAc buffer solution (0.1 M, pH 7) were incubated at 37 °C for 30 min. Subsequently, 1 mL of the NaOAc buffer solution (pH 3.8), 200 μL of R-Co₃O₄ (0.4 mg/mL), and 200 μL of the TMB (1 mM) solution were added to the aforementioned solution. The mixture was

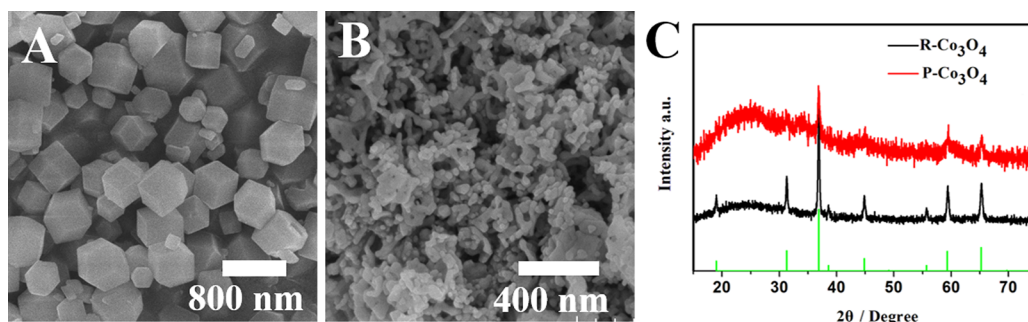


Figure 1. SEM images of ZIF-67 (A) and P-Co₃O₄ (B). PXRD patterns of P-Co₃O₄ and R-Co₃O₄ (C).

incubated at 37 °C for 30 min and was further used for performing the absorption spectroscopy measurement.

2.7. DFT Calculations for the H₂O₂ Adsorption on P-Co₃O₄ and R-Co₃O₄. The DMol3 package from Materials Studio was used for performing all the calculations.³¹ The Perdew–Burke–Ernzerh (PBE) functional of the generalized gradient approximation was selected for performing the unrestricted spin-polarized DFT calculations. The DND basis set (version 3.5) was used, and all the core electrons of Co were treated using the effective core potentials. The real space cutoff radius was set to 4.1 Å throughout the study. Furthermore, the Brillouin zone was integrated using the Monkhorst–Pack scheme with a grid of $2 \times 2 \times 1$. The convergence criteria for structure optimization and energy calculation were set to (a) an SCF tolerance of 1×10^{-5} hartree, (b) an energy tolerance of 2×10^{-5} hartree, (c) a maximum force tolerance of 4×10^{-3} hartree/Å, and (d) a maximum displacement tolerance of 5×10^{-3} Å.

The geometrically optimized lattice parameters (primitive cell) of Co₃O₄ ($a = b = c = 5.76$ Å and $\alpha = \beta = \gamma = 60^\circ$) are in good agreement with the experimental values ($a = b = c = 5.72$ Å and $\alpha = \beta = \gamma = 60^\circ$), indicating the credibility of the calculations. On the basis of the XRD, HRTEM, and selected-area electron diffraction (SAED) results, the calculation focuses on the cubic Co₃O₄(113) surface. After cleaving the (113) surface of the optimized crystal structure of Co₃O₄ with a thickness of 8.17 Å, a vacuum zone with a thickness of 20 Å was added to eliminate any interactions between the adjacent slabs. The adsorption energies (E_{ads}) of H₂O₂ on the cubic Co₃O₄(113) surface are calculated from the relation $E_{\text{ads}} = E_{(\text{molecule+surface})} - (E_{\text{molecule}} + E_{\text{surface}})$, where $E_{(\text{molecule+surface})}$, E_{molecule} , and E_{surface} represent the energies of H₂O₂ on the cubic Co₃O₄(113) surface, the isolated H₂O₂ molecule, and the Co₃O₄(113) surface, respectively. Furthermore, two possible adsorption sites and orientations of H₂O₂ [P-Co₃O₄(113)_H₂O₂-1 and P-Co₃O₄(113)_H₂O₂-2] were investigated near the Co³⁺ or Co²⁺ atoms on the P-Co₃O₄(113) surface. For the adsorption of H₂O₂ on the R-Co₃O₄(113) surface, one center O atom of P-Co₃O₄(113)_H₂O₂-1 and P-Co₃O₄(113)_H₂O₂-2 were deleted to model the O vacancy, resulting in R-Co₃O₄(113)_H₂O₂-1 and R-Co₃O₄(113)_H₂O₂-2 models.

3. RESULTS AND DISCUSSION

3.1. Synthesis and Characterization of R-Co₃O₄. The synthetic route to obtain R-Co₃O₄ is illustrated in Scheme 1. ZIF-67 is prepared by hydrothermal synthesis using 2-methylimidazole and Co(NO₃)₂·6H₂O in methanol. The XRD pattern confirms the phase purity of the as-prepared ZIF-67 (Figure S1). The SEM images indicate that ZIF-67 presents a uniform and perfect polyhedral morphology with smooth surfaces (Figure 1A).

Subsequently, ZIF-67 was used as a sacrificial precursor to prepare P-Co₃O₄ under an oxygen atmosphere at 500 °C. The SEM images show that the polyhedral morphology transforms into that of a coral-like nanocomposite after pyrolytic treatment, which is most likely due to the transformation of

the organic ligands into carbonaceous materials (Figure 1B). In the XRD pattern, the peaks observed at 18.98, 31.32, 36.86, 38.62, 44.85, 59.29, and 65.31° can be assigned to the (111), (022), (113), (222), (004), (115), and (044) planes of cubic Co₃O₄ (JCPDS card No. 42-1467), respectively. No other peaks can be observed, indicating the high purity of the prepared P-Co₃O₄ (Figure 1C).

After treatment with NaBH₄, diffraction peaks can be observed at 36.86, 44.85, 59.29, and 65.31°, indicating that the cubic Co₃O₄ crystal structure was preserved upon treatment with NaBH₄ to prepare R-Co₃O₄ (Figure 1C). The TEM and SEM images indicate that R-Co₃O₄ presents a morphology featuring flowerlike structures of uniform sizes and petal widths of ~14 nm (Figure 2A,B). The HRTEM image of the R-Co₃O₄

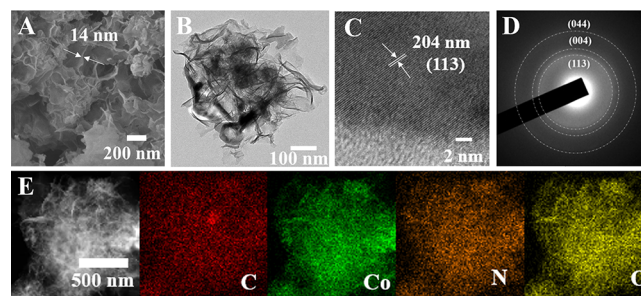


Figure 2. (A) SEM and (B) TEM images, (C) HRTEM image, (D) SAED patterns, and (E) EDS mapping images of R-Co₃O₄.

sample is presented in Figure 2C. A lattice spacing of 2.04 nm can be assigned to the (113) plane of cubic Co₃O₄. The SAED pattern presents polycrystalline diffraction rings corresponding to the (113), (004), and (044) planes of cubic Co₃O₄, corroborating the XRD results (Figure 2D). The R-Co₃O₄ components and structure were investigated by scanning TEM energy-dispersive X-ray spectroscopy (STEM–EDS) elemental mapping (Figure 2E), revealing the uniform distribution of the Co, C, N, and O elements in the sample. Furthermore, XPS was employed for elucidating the surface chemical states of Co in P-Co₃O₄ and R-Co₃O₄ (Figure 3A). Compared to P-Co₃O₄, R-Co₃O₄ presents two additional peaks at 803.8 and 786.7 eV, which can be attributed to Co²⁺, indicating the transformation of Co³⁺ to Co²⁺ and the formation of oxygen vacancies in R-Co₃O₄.^{32,33} As depicted in Figure 3B, the ESR spectra also confirm the existence of oxygen vacancies in R-Co₃O₄, which is in accordance with previous reports.^{34,35} N₂ adsorption–desorption analysis was conducted at 77 K to investigate the porosity of R-Co₃O₄ (Figure 3C). R-Co₃O₄ shows a typical type-IV adsorption–desorption isotherm with a BET specific surface area of 64.7 m²/g and pore sizes of 2.4–3.8 nm (Figure

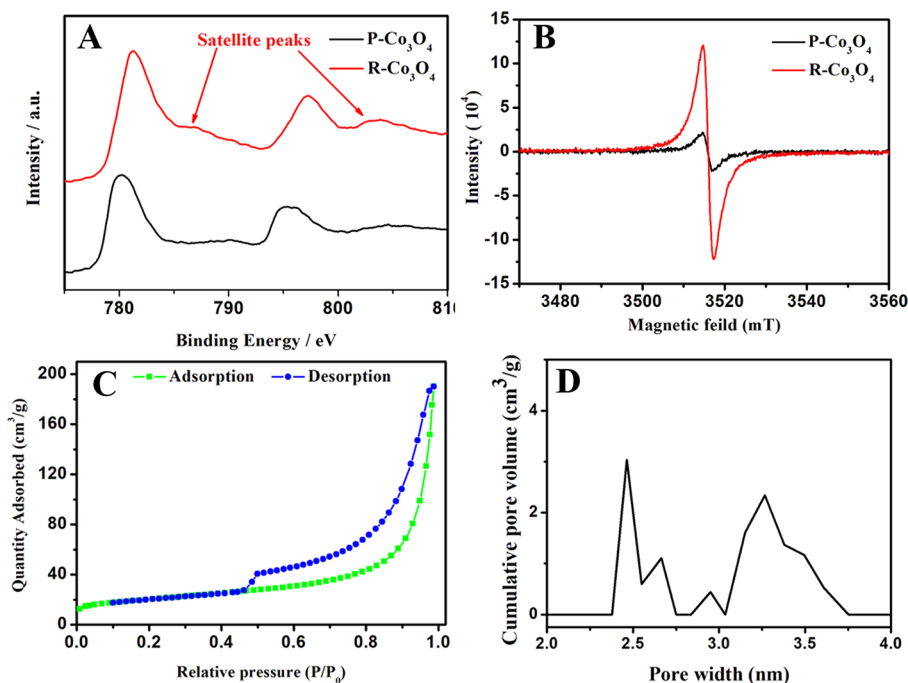


Figure 3. (A) XPS spectra and (B) ESR spectra of P-Co₃O₄ and R-Co₃O₄. (C) N₂ adsorption–desorption isotherms of R-Co₃O₄; (D) pore size distribution curve from the BJH model.

3D). The large surface area and abundant mesoporous channels favor the rapid mass/charge transportation and provide ample catalytically active sites for the catalytic process.^{12,36,37}

3.2. Characterization of the Peroxidase-like Activity of R-Co₃O₄. The peroxidase-like activities of P-Co₃O₄ and R-Co₃O₄ were comparatively studied by the catalytic oxidation of TMB to blue-colored oxTMB in the presence of H₂O₂ (Figures 4A and S2). The reaction system presents no color change when the catalyst is absent. However, a blue color can be observed in the presence of P-Co₃O₄ and R-Co₃O₄. The reaction system with R-Co₃O₄ denotes darker blue color. Accordingly, the UV–vis absorbance spectra denote that R-

Co₃O₄ exhibits an improved peroxidase-like activity when compared with that exhibited by P-Co₃O₄, indicating that the introduction of oxygen vacancies into Co₃O₄ is an effective strategy to enhance its peroxidase-like activity.

As is known, the peroxidase-like activities of the natural and artificial enzymes are dependent on the external conditions. Therefore, several factors, such as the pH, temperature, and concentration of H₂O₂ are optimized (Figure 4B–D). The peroxidase-like activities of R-Co₃O₄ and P-Co₃O₄ increase with increasing pH from 3.5 to 4. However, the peroxidase-like activities decrease upon a further increase in the pH of the reaction system (Figure 4B). The temperature of the reaction system is also influential with respect to the peroxidase-like activity. Both the systems exhibit their optimal peroxidase-like activity at 45 °C. However, unlike P-Co₃O₄, R-Co₃O₄ exhibits considerable peroxidase-like activity over the entire tested temperature range (Figure 4C). The peroxidase-like activity of P-Co₃O₄ quickly improves with the concentration of H₂O₂ and reaches its maximum at the H₂O₂ concentration of 6 mM. Thereafter, the peroxidase-like activity rapidly declines. In case of R-Co₃O₄, the effect of H₂O₂ concentration on the peroxidase-like activity is not significant, and the optimal H₂O₂ concentration is 17 mM (Figure 4D). It is worth pointing out that R-Co₃O₄ has better catalytic activity over a wider range of pH, temperature, and concentration of H₂O₂ than that of P-Co₃O₄, which can be attributed to its unique nanostructure where its active centers are embedded in porous carbon. The subsequent colorimetric assays were employed under individual optimal experimental conditions.

3.3. Steady-State Kinetic Analysis of R-Co₃O₄. A steady-state kinetic analysis of R-Co₃O₄ toward substrates was performed to investigate the kinetic mechanism of the peroxidase-like activity of R-Co₃O₄ (Figure 5). Typically, the Michaelis–Menten equations can be obtained by varying the concentration of a substrate while maintaining the remaining conditions unchanged. The Michaelis–Menten constant (K_m)

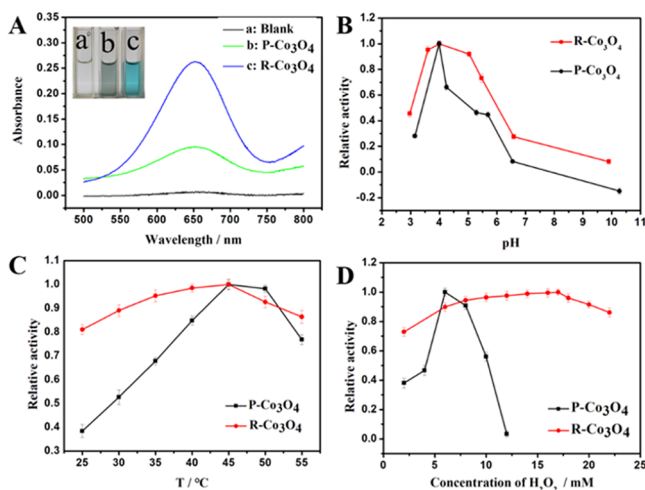


Figure 4. (A) Photo of color changes and UV–vis spectra of TMB in different reaction system; (B) pH-dependent, (C) temperature-dependent, and (D) H₂O₂-concentration-dependent peroxidase-like activity of P-Co₃O₄ and R-Co₃O₄.

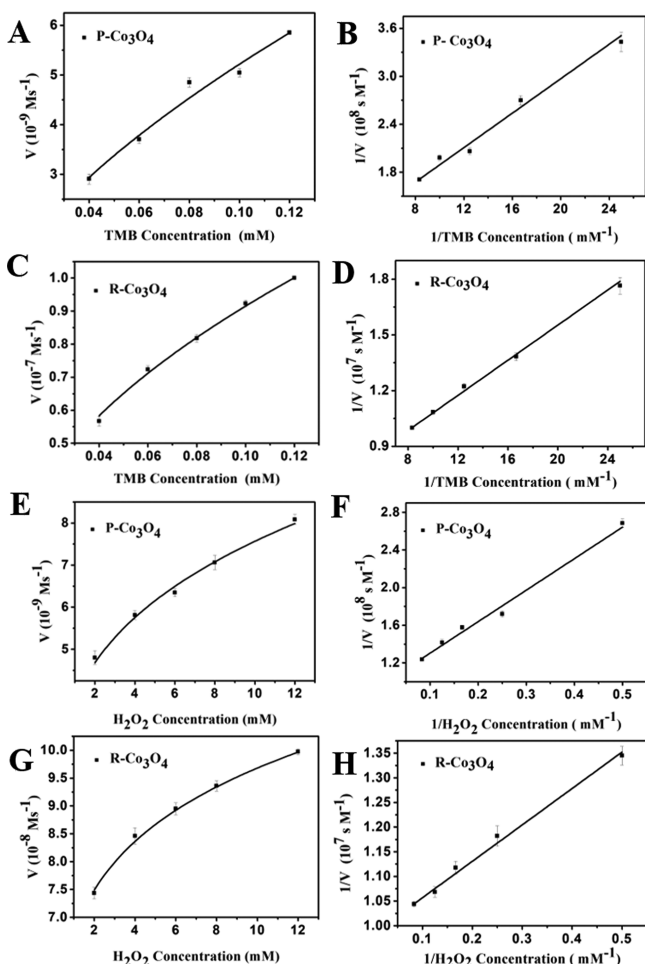


Figure 5. Steady-state kinetic assay of P-Co₃O₄ and R-Co₃O₄.

and maximum initial velocity ($V_{\max}$) can be obtained based on the double reciprocal of the Michaelis–Menten equation (Table 1). The K_m value of R-Co₃O₄ obtained using TMB as

Table 1. Comparison of Kinetic Parameters of P-Co₃O₄, R-Co₃O₄, and Recently Reported Artificial Enzymes

catalyst	K_m [mM]		$V_{\max}$ [10^{-8} s ⁻¹]		ref
	H ₂ O ₂	TMB	H ₂ O ₂	TMB	
Au/Co ₃ O ₄ -CeOx NCs	0.27	0.12	0.39	0.86	19
HRP	3.70	0.43	10.00	8.71	38
Co ₃ O ₄ @CeO ₂	7.09	0.14	43.3	41.3	39
Co ₃ O ₄ -MMT	20.49	0.0021	34.23	1.75	40
Co ₃ O ₄ NPs	140.07	0.037	12.10	6.27	41
P-Co ₃ O ₄	3.43	0.12	1.03	1.14	this work
R-Co ₃ O ₄	0.74	0.07	10.13	15.80	this work

the substrate is only slightly lower than that of P-Co₃O₄ (Figure 5A–D). However, the K_m value of R-Co₃O₄ obtained using H₂O₂ as the substrate is significantly lower than that of P-Co₃O₄, indicating that the introduction of oxygen vacancies into Co₃O₄ effectively improves its affinity to H₂O₂ (Figure 5E–H). Furthermore, the K_m values of R-Co₃O₄ obtained using H₂O₂ and TMB as the substrates are significantly lower than those of HRP and other artificial enzymes, demonstrating its superior affinity to substrates (Table 1).

3.4. Durability Analysis. As is widely known, natural enzymes show low stabilities, severely hindering their application. The samples were incubated at pH 3–10 and temperatures of 0–90 °C for 24 h to assess the peroxidase-like stability of R-Co₃O₄. Subsequently, their peroxidase-like activities were monitored under optimal conditions by UV–vis spectrophotometry. As shown in Figure 6A, R-Co₃O₄

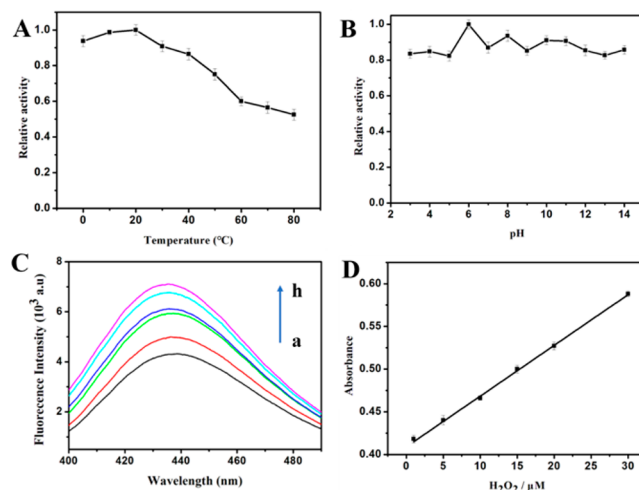


Figure 6. Durability of R-Co₃O₄ after incubating at various temperature (A) and pH values (B) for 24 h; the effect of R-Co₃O₄ concentrations on the formation of hydroxyl radical with terephthalic acid as a fluorescence probe (C). (a–f) 0, 40, 80, 120, 160, 180 μg/mL; linear calibration plot for H₂O₂ detection using R-Co₃O₄ as artificial enzyme (D).

retains ~54% of its initial catalytic activity at 80 °C, whereas the catalytic activity of HRP clearly decreases to ~10%,¹⁸ demonstrating the excellent heat resistance of R-Co₃O₄. Figure 6B indicates that R-Co₃O₄ retains the considerable catalytic activity when the pH is 3–14. However, the decomposition of HRP in strong acidic or alkaline solutions results in declining catalytic activity.³⁸ The excellent durability of R-Co₃O₄ in case of harsh environments can be attributed to its unique nanostructure, where its active centers are embedded in porous carbon.

3.5. Mechanism Analysis. The intrinsic peroxidase-like activity of P-Co₃O₄ and R-Co₃O₄ may be attributed to their catalytic activation of H₂O₂ through electron transfer to form ·OH radicals, which oxidizes TMB to blue $oxTMB$. Terephthalic acid can react with the ·OH radicals to generate the highly fluorescent 2-hydroxy terephthalic acid.⁴² Thus, terephthalic acid can be used as a fluorescent probe to detect the existence of the ·OH radicals. As denoted in Figures 6C and S3, the fluorescence intensity increases with increasing concentration for both P-Co₃O₄ and R-Co₃O₄, indicating that the amounts of ·OH radicals increase accordingly. This confirms that P-Co₃O₄ and R-Co₃O₄ efficiently catalyze the H₂O₂ decomposition into ·OH radicals, resulting in their peroxidase-like activity.

3.6. Colorimetric Detection of H₂O₂ and Glucose Based on R-Co₃O₄. On the basis of the excellent peroxidase-like activity of R-Co₃O₄, a simple and convenient colorimetric assay has been developed for the quantitative detection of H₂O₂. Figures S4 and 6D show that the UV–vis absorption intensity of the R-Co₃O₄-based system increases linearly with

the H_2O_2 concentration in 1–30 μM with a detection limit of 4.3×10^{-7} mol/L ($S/N = 3$).

Glucose oxidase (GOx) promotes the catalytic oxidation of glucose to produce H_2O_2 . Therefore, the $\text{R-Co}_3\text{O}_4$ -based system can be used for the quantitative detection of glucose in the presence of GOx. Figure S5 shows a typical glucose concentration response curve for $\text{R-Co}_3\text{O}_4$ from 1 to 20 μM with a detection limit of 3.2×10^{-7} mol/L, which is superior to that of several previously reported artificial enzymes (Table 2).

Table 2. Comparison of Linear Range and Detection Limit of $\text{P-Co}_3\text{O}_4$, $\text{R-Co}_3\text{O}_4$, and Recently Reported Artificial Enzymes

catalyst	linear range		limit of detection (LOD)		ref
	H_2O_2 (μM)	glucose (μM)	H_2O_2 (μM)	glucose (μM)	
Co_3O_4 -MMT	10–100		8.7		40
Co_3O_4 NPs	50–25000	10–10000	10	5	41
PDI- Co_3O_4	3–60	5–100	2.37	2.77	43
CoNPs/MC	0.1–20	0.25–30	0.077	0.156	44
$\text{Fe}_3\text{O}_4@\text{MS NPs}$	0.1–100	10–500	1	4	45
$\text{P-Co}_3\text{O}_4$	1–30	10–30	0.77	0.69	this work
$\text{R-Co}_3\text{O}_4$	1–30	1–20	0.43	0.32	this work

Furthermore, the selectivity of the colorimetric detection of glucose based on the proposed $\text{R-Co}_3\text{O}_4$ -based system was investigated via control experiments. Several typical common sugars and glucose were comparatively analyzed, and the results are presented in Figure 7. The absorption intensity of

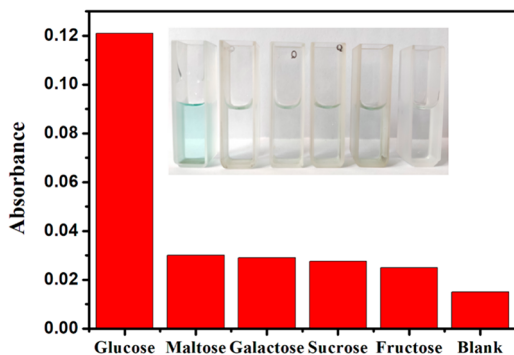


Figure 7. Selectivity of colorimetric detection of glucose based on the proposed $\text{R-Co}_3\text{O}_4$ based system.

the $\text{R-Co}_3\text{O}_4$ -based system is clearly higher than those of the control samples, indicating that the proposed $\text{R-Co}_3\text{O}_4$ -based system exhibits high selectivity for glucose.

The quantitative detection of glucose in human serum was conducted using the proposed $\text{R-Co}_3\text{O}_4$ -based system to investigate the application of this colorimetric system to complex biological matrices. Before the test, the serum samples were diluted to ensure that the concentrations of glucose in the samples fell within the linear range for glucose. Recoveries of 95.6–107.6% are observed (Table S1), indicating that the proposed $\text{R-Co}_3\text{O}_4$ -based system could be effectively used for the highly sensitive colorimetric detection of glucose in serum.

3.7. DFT Calculations. The structures of the H_2O_2 adsorption configurations in the cases of the $\text{P-Co}_3\text{O}_4(113)$ surface and the $\text{R-Co}_3\text{O}_4(113)$ surface [$\text{P-Co}_3\text{O}_4(113)\text{-H}_2\text{O}_2\text{-1}$, $\text{P-Co}_3\text{O}_4(113)\text{-H}_2\text{O}_2\text{-2}$, $\text{R-Co}_3\text{O}_4(113)\text{-H}_2\text{O}_2\text{-1}$, and $\text{R-Co}_3\text{O}_4(113)\text{-H}_2\text{O}_2\text{-2}$] are presented in Figure 8. The

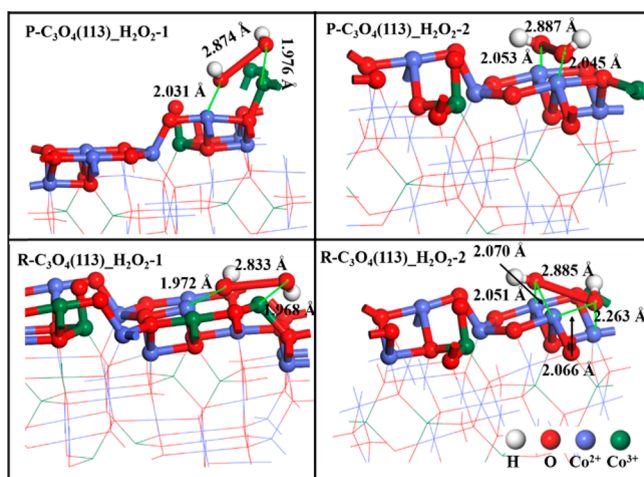


Figure 8. Structures of the H_2O_2 adsorption configurations on P- and $\text{R-Co}_3\text{O}_4$ (113) surfaces.

optimized parameters for H_2O_2 adsorption on $\text{P-Co}_3\text{O}_4(113)$ surface and the $\text{R-Co}_3\text{O}_4(113)$ surface are summarized in Table S2. In all the configurations, H_2O_2 is chemically adsorbed on the top of surface with both the O atoms of H_2O_2 being bonded with one or two Co atoms. The Co–O bond lengths are 2.031/1.976, 2.053/2.045, 1.972/1.968, and 2.051/2.070/2.066/2.263 Å for $\text{P-Co}_3\text{O}_4(113)\text{-H}_2\text{O}_2\text{-1}$, $\text{P-Co}_3\text{O}_4(113)\text{-H}_2\text{O}_2\text{-2}$, $\text{R-Co}_3\text{O}_4(113)\text{-H}_2\text{O}_2\text{-1}$, and $\text{R-Co}_3\text{O}_4(113)\text{-H}_2\text{O}_2\text{-2}$, respectively. The corresponding adsorption energies are −48.1, −54.7, −63.4, and −70.3 kcal/mol. These results indicate that $\text{R-Co}_3\text{O}_4$ with its oxygen vacancies exhibits a considerably larger adsorption energy when compared with that exhibited by $\text{P-Co}_3\text{O}_4$, containing no oxygen vacancies. Furthermore, the O–O bond in the H_2O_2 molecule is elongated to lengths of 2.874, 2.887, 2.833, and 2.885 Å, respectively, considerably longer than the O–O bond (1.470 Å) of an isolated H_2O_2 molecule. These results indicate that the H_2O_2 molecules are activated and dissociated after adsorption. Therefore, the introduction of oxygen vacancies into Co_3O_4 enhances the adsorption intensity of H_2O_2 and facilitates the decomposition of H_2O_2 to produce $\cdot\text{OH}$ radicals.

4. CONCLUSIONS

In summary, $\text{R-Co}_3\text{O}_4$ has been synthesized via a step-by-step procedure using ZIF-67 as the precursor. Compared to $\text{P-Co}_3\text{O}_4$, $\text{R-Co}_3\text{O}_4$ exhibits remarkably enhanced peroxidase-like activity. The DFT calculations revealed that the introduction of oxygen vacancies into Co_3O_4 enhances the adsorption of H_2O_2 and facilitates its decomposition to produce $\cdot\text{OH}$ radicals, resulting in an improved peroxidase-like activity. A highly sensitive and selective colorimetric assay has been established to detect H_2O_2 and glucose on the basis of the excellent peroxidase-like activity of $\text{R-Co}_3\text{O}_4$.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.inorgchem.9b03512>.

Powder XRD pattern and the simulated one from the single-crystal diffraction data of ZIF-67, time-dependent absorbance changes at 652 nm of TMB in different reaction systems, effect of P-Co₃O₄ on the formation of hydroxyl radical using terephthalic acid as fluorescence probe, UV-vis absorbance spectra changes versus H₂O₂ concentrations of the R-Co₃O₄ based system, linear calibration plot for glucose detection using R-Co₃O₄ as artificial enzyme, analytical results of serum samples by the proposed method, and optimized parameters for H₂O₂ adsorption on P-Co₃O₄(113) surface and the R-Co₃O₄(113) surface (PDF)

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Notes

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